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To
Hiring Manager,
ExxonMobil,
22777 Springwoods Village Parkway, Spring, TX 77389

September 28, 2025

Sub: Application for Geophysics Internship

Dear Hiring Manager,

I am writing to express my enthusiastic interest in a geophysics internship at ExxonMobil. As a third-year Geophysics Ph.D. student at the Colorado School of Mines, my research has been sponsored by ExxonMobil, which has given me a deep appreciation for your company's innovative work. I believe my skills are exceptionally well-aligned with the challenges and opportunities at ExxonMobil.

My Ph.D. research on using Distributed Fiber-Optic Sensing (DFOS) for hydraulic fracture characterization directly matches the needs of a leading energy company like ExxonMobil. On the ExxonMobil-sponsored "Project Mariner," I developed a complete low-frequency DAS (LF-DAS) data processing workflow and built a coupled pressure-strain numerical model using MOOSE to perform history matching against field data. This experience solidified my ability to analyze large datasets and model subsurface fluid dynamics.

To support my research, I developed fiberRIS, an open-source Python package designed for the history matching and interpretation of DAS data, hence to characterize subsurface reservoirs. This project demonstrates my advanced proficiency in Python and my passion for creating practical tools for data analysis. Furthermore, my academic coursework in Advanced Machine Learning and Advanced Data Science has provided me with a strong foundation to explore the scientific machine learning approaches that are becoming central to the energy industry.

I am very excited by the prospect of contributing to your team. The opportunity to apply my expertise in DAS analysis, numerical modeling, and Python development to advance ExxonMobil's projects is incredibly compelling.

Thank you for your consideration. I have attached my resume and welcome the opportunity to discuss how my experience can benefit ExxonMobil.

Sincerely,

Shenyao Jin